

Module 04 Lab - Exploratory Data

Analysis Objective: To learn how to explore a dataset to find patterns, anomalies, and insights before modeling. EDA is like being a detective; you are looking for clues in the data that will help you build a better model. **This lab is fully coded.** Your task is to run each cell, read the detailed explanations, understand the purpose of each visualization, and then complete the experimentation section.

Part 1: Setup and Data Loading**What is Exploratory Data Analysis (EDA)?**EDA is the process of using summary statistics and visualizations to understand a dataset's main characteristics. Before you can build a model, you need to understand your data. What stories does it tell? Are there errors or missing values? Are there strong relationships between variables? EDA helps answer these questions. We will use the famous Titanic dataset for this lab. It contains information about passengers and, crucially, whether they survived the disaster.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
# Seaborn is a library built on top of Matplotlib that makes creating b

# Load the dataset directly from a URL
df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')

print("---- First 5 Rows ----")
print(df.head())

print("---- Basic Info ----")
df.info()
# .info() is a great first command. It tells us the column names, how many non-null values, and the data types.
# Notice that 'Age' and 'Cabin' have missing values!
```

```
---- First 5 Rows ----
   PassengerId  Survived  Pclass
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3
```

```

0                                     Name      Sex  Age  Sib
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0
2                                     Heikkinen, Miss. Laina  female  26.0
3      Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0
4                                     Allen, Mr. William Henry  male  35.0

   Parch      Ticket    Fare Cabin Embarked
0      0         A/5 21171    7.2500   NaN      S
1      0          PC 17599   71.2833   C85      C
2      0  STON/O2. 3101282    7.9250   NaN      S
3      0          113803   53.1000  C123      S
4      0          373450    8.0500   NaN      S

--- Basic Info ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

```

Part 2: Descriptive Statistics Let's start by getting a high-level numerical summary of the data. The `.describe()`

- method is perfect for this. It calculates statistics like mean, standard deviation, min, and max for the numerical columns.

```
# Get summary statistics for numerical columns
print("--- Descriptive Statistics ---")
print(df.describe())
#age has fewer insight because some passengers age could be missing.
print("--- Key Insights from Statistics ---")
print(f"The average age of a passenger was {df['Age'].mean():.1f} years.")
print(f"The overall survival rate was {df['Survived'].mean():.1%}.")
print(f"Fares ranged from ${df['Fare'].min()} to a whopping ${df['Fare'].max()}")
```

--- Descriptive Statistics ---

	PassengerId	Survived	Pclass	Age	SibSp	\
count	891.000000	891.000000	891.000000	714.000000	891.000000	
mean	446.000000	0.383838	2.308642	29.699118	0.523008	
std	257.353842	0.486592	0.836071	14.526497	1.102743	
min	1.000000	0.000000	1.000000	0.420000	0.000000	
25%	223.500000	0.000000	2.000000	20.125000	0.000000	
50%	446.000000	0.000000	3.000000	28.000000	0.000000	
75%	668.500000	1.000000	3.000000	38.000000	1.000000	
max	891.000000	1.000000	3.000000	80.000000	8.000000	

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

--- Key Insights from Statistics ---

The average age of a passenger was 29.7 years.

The overall survival rate was 38.4%.

Fares ranged from \$0.0 to a whopping \$512.3292.

Part 3: Visual EDA - Telling Stories with Plots Numbers are great, but plots make patterns and relationships immediately obvious. The goal of visual EDA is to turn data into insights.

A Note on Plotting Libraries:

- * **Matplotlib:**

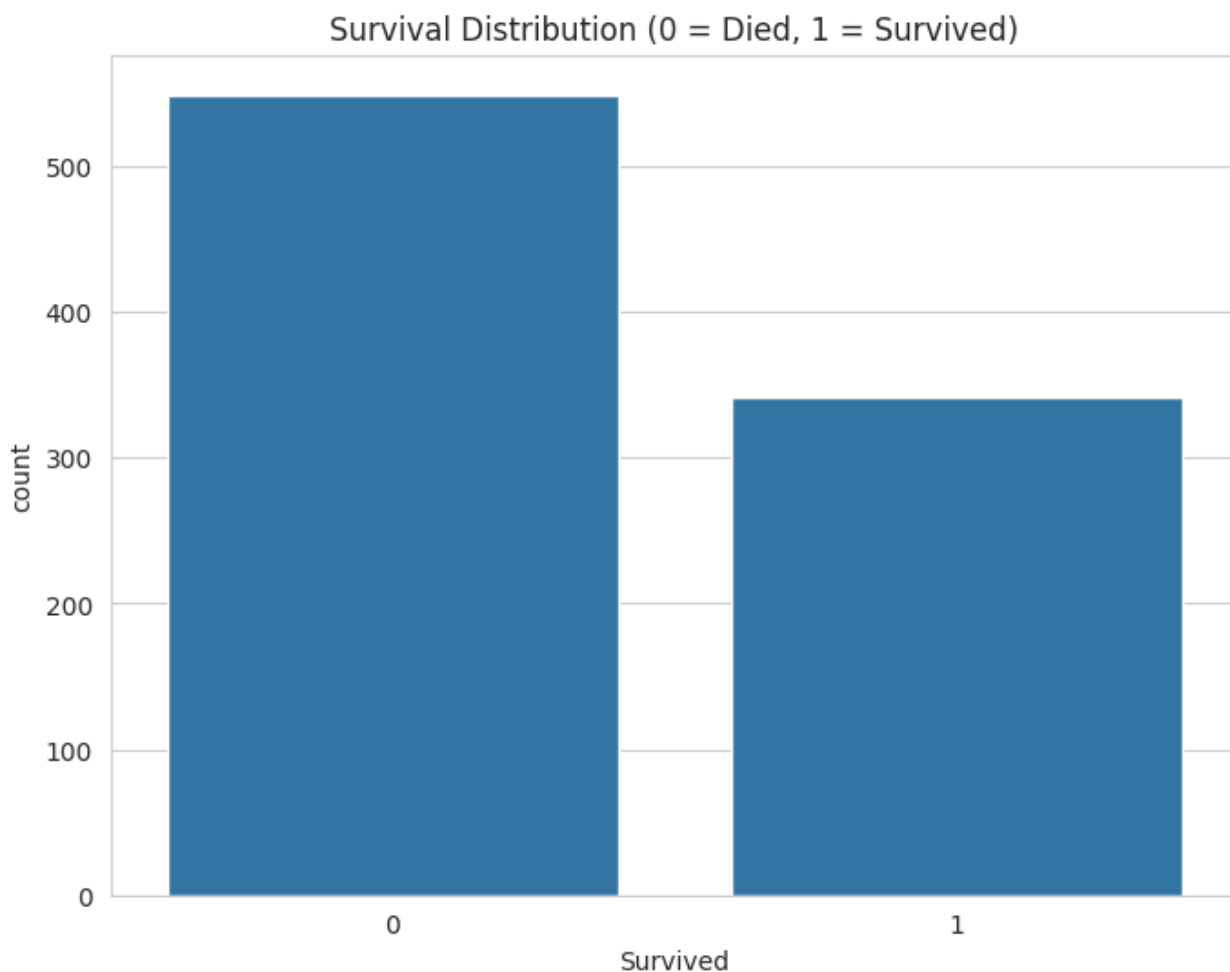
- ✓ The foundational library, gives you full control over everything.
- * **Seaborn:** Built on Matplotlib, it provides a simpler, high-level interface for creating common statistical plots. We will use Seaborn for its ease of use and attractive defaults.

Visualization 1: How many survived? A simple `countplot` is the best

- ✓ way to see the distribution of a categorical variable, like our target `Survived`.

```
sns.set_style('whitegrid')
# Sets a nice visual style for our plots
plt.figure(figsize=(8, 6))
sns.countplot(x='Survived', data=df)
plt.title('Survival Distribution (0 = Died, 1 = Survived)')
plt.show()

# There's a possible imbalance dataset can make some outcomes to not occur
print("Insight: Far more people died than survived. This is an example of an imbalance dataset")
```



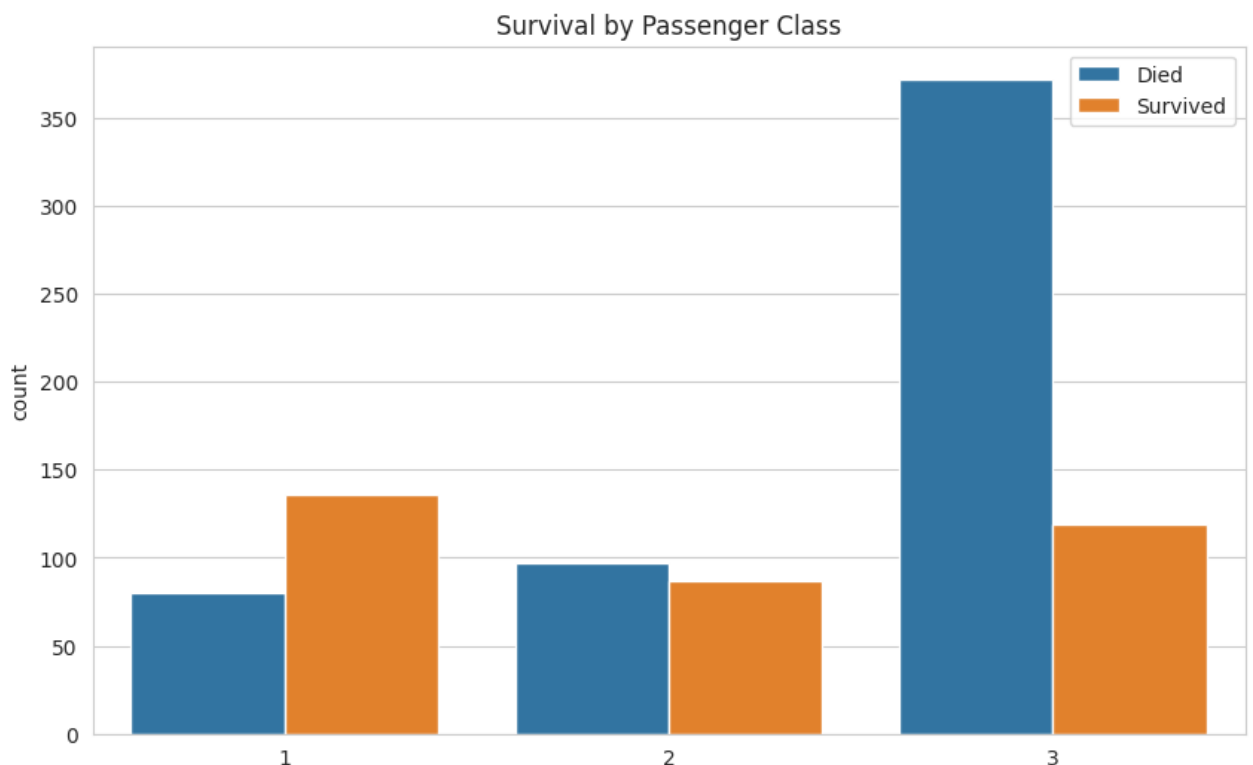
Insight: Far more people died than survived. This is an example of an imbalance dataset

Visualization 2: Does passenger class matter for survival? Now we want to see if there's a relationship between two variables. We can use the `hue` parameter in `countplot` to split the bars by another category.

```
plt.figure(figsize=(10, 6))

sns.countplot(
    x='Pclass',
    hue='Survived',
    data=df
)

plt.title('Survival by Passenger Class')
plt.xlabel('passenger class')
plt.ylabel('count')
plt.legend(['Died', 'Survived'])
plt.show()
#more first class passangers survived than died but 3rd class has high
#2nd class was more balanced between the two.
print("Insight: This is a very strong pattern. 1st class passengers ha
```



passenger class

Insight: This is a very strong pattern. 1st class passengers had a much

v

Visualization 3: What about gender? The 'women and children first' mantra is famous. Let's see if the data supports it.

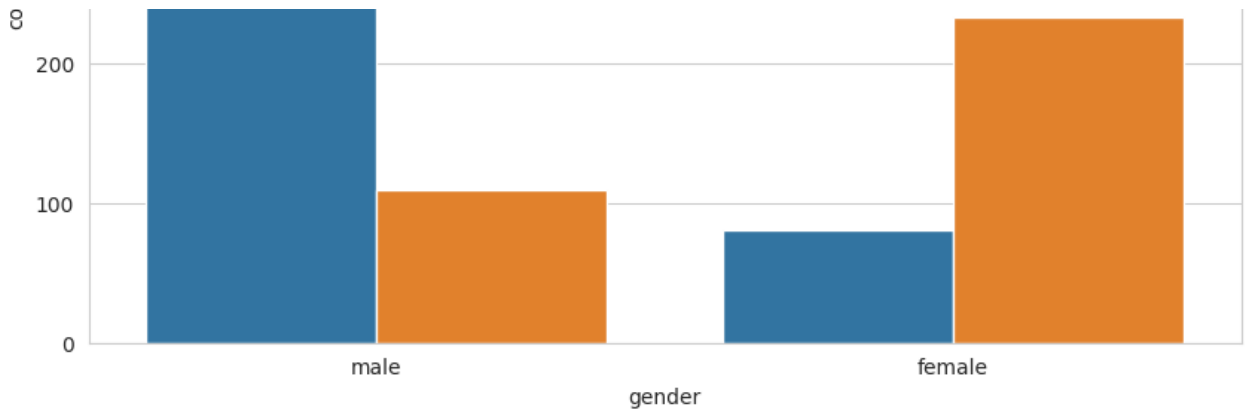
```
plt.figure(figsize=(10, 6))

sns.countplot(
    x='Sex',
    hue='Survived',
    data=df
)

plt.title('Survival by Gender')
x_label = plt.xlabel('gender')
y_label = plt.ylabel('count')
plt.legend(['Died', 'Survived'])
plt.show()

#Male passengers did not survive
#Female passengers have the highest survival rate by a good count.
print("Insight: The pattern is undeniable. A much higher proportion of
```





Insight: The pattern is undeniable. A much higher proportion of females

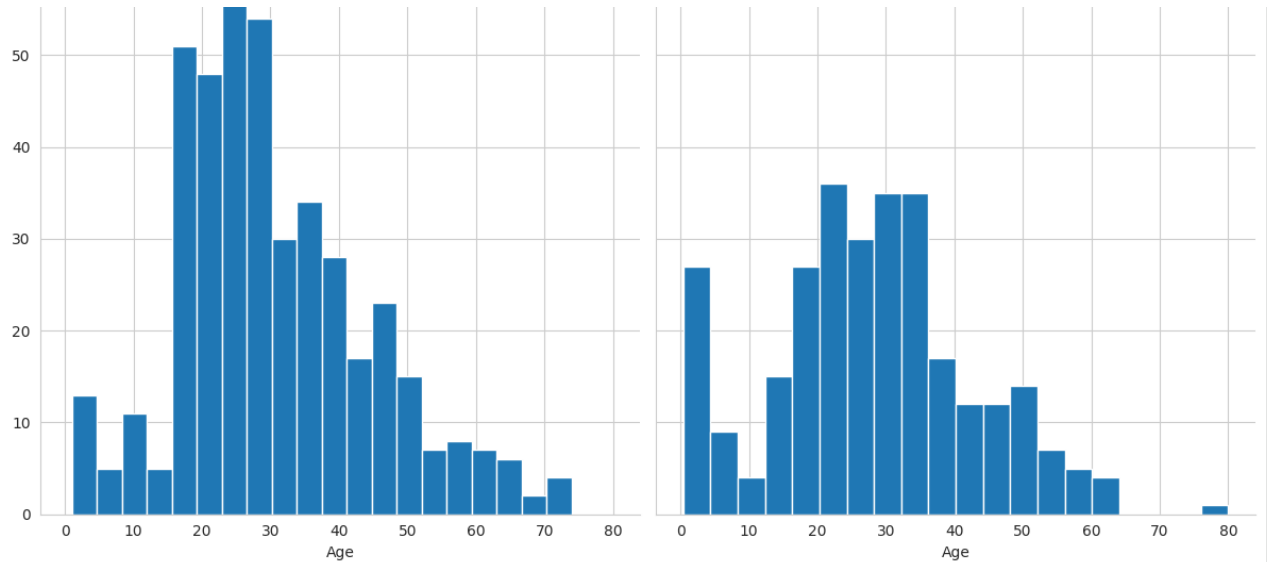
Visualization 4: How does age play a role? For a continuous variable like **Age**, a histogram is a great way to see its distribution.

```
# A FacetGrid allows us to create multiple plots side-by-side to compare
# Here, we create one histogram for passengers who died (col='Survived')

g = sns.FacetGrid(df,
                    col='Survived',
                    height=6
)
g.map(plt.hist,
      'Age',
      bins=20
)
#survive 0 = Died
#survive 1 = Survive

plt.show()
print("Insight: The age distribution for those who did not survive is
```





Insight: The age distribution for those who did not survive is centered around age 25. For those who survived, there is a noticeable spike for young children (ages 0-10).

Part 4: Student Experimentation **Instructions:** Create your own visualizations to explore other relationships in the data.

Experiment 1: Port of Embarkation1. The **Embarked** column tells you where a passenger boarded the ship (C = Cherbourg, Q =

- Queenstown, S = Southampton).2. Create a **countplot** to see how survival rates differed by the port of embarkation. Does where they boarded seem to be related to their survival?

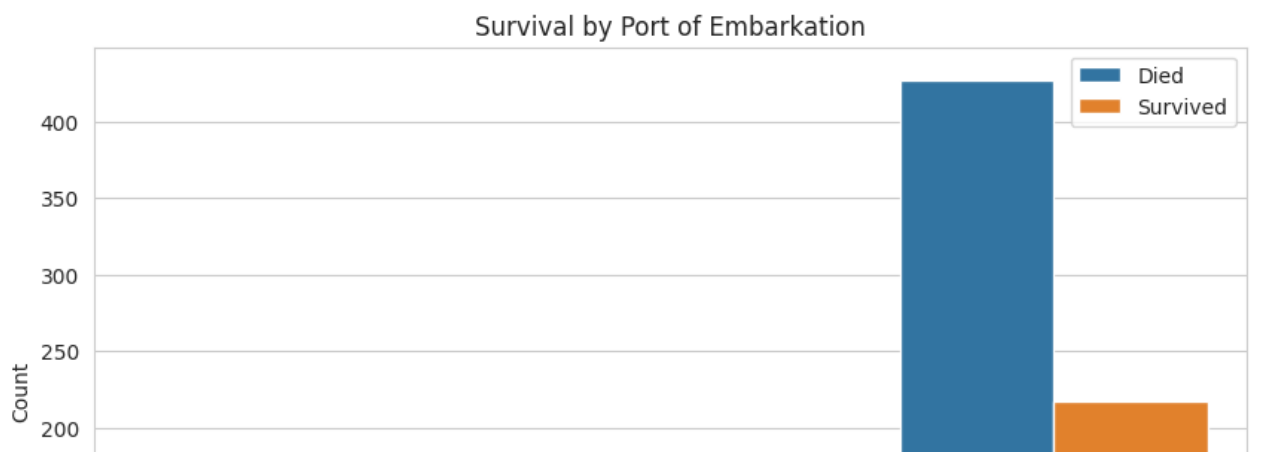
```
# --- ENTER YOUR CODE HERE ---

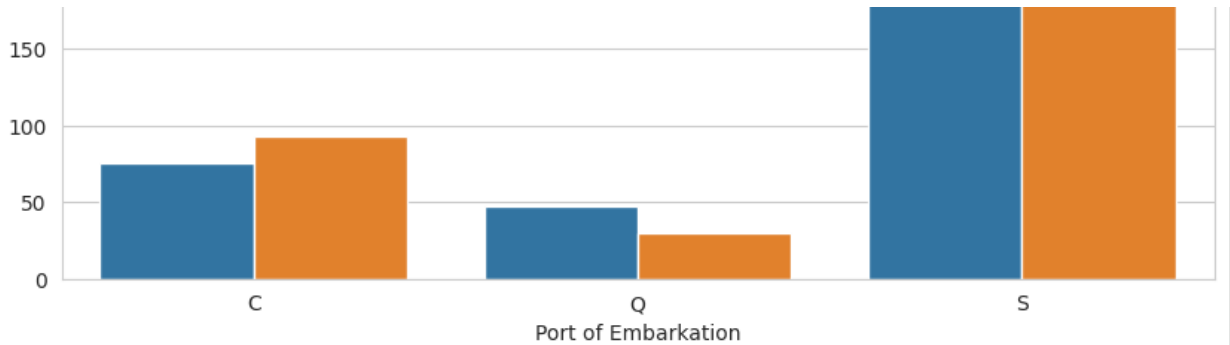
plt.figure(figsize=(10, 6))

sns.countplot(
    x = 'Embarked',
    hue = 'Survived',
    data = df,
    order = ["C", "Q", "S"]
)

plt.title('Survival by Port of Embarkation')
plt.xlabel('Port of Embarkation')
plt.ylabel('Count')
plt.legend(['Died', 'Survived'])
plt.show()

survival_by_port = df.groupby('Embarked')['Survived'].mean() * 100
print("Survival Rate by Port of Embarkation1:")
print(survival_by_port)
#embarkation port has a connection to the survivalists, but the port co
print("\n Insight: Passangers that boarded at Cherbourg had the highest
```





Survival Rate by Port of Embarkation1:

Embarked

C 55.357143

Q 38.961039

S 33.695652

Name: Survived, dtype: float64

Insight: Passangers that boarded at Cherbourg had the highest survival rate, while passangers that boarded the Southampton had the lowest sur

- Experiment 2: Fare vs. Survival1. **Fare** is a continuous numerical variable.2. A **boxplot** or **violinplot** is a great way to see the distribution of fares for those who survived vs. those who didn't.3. Create one of these plots to compare the **Fare** distribution by **Survived**. What does it tell you?

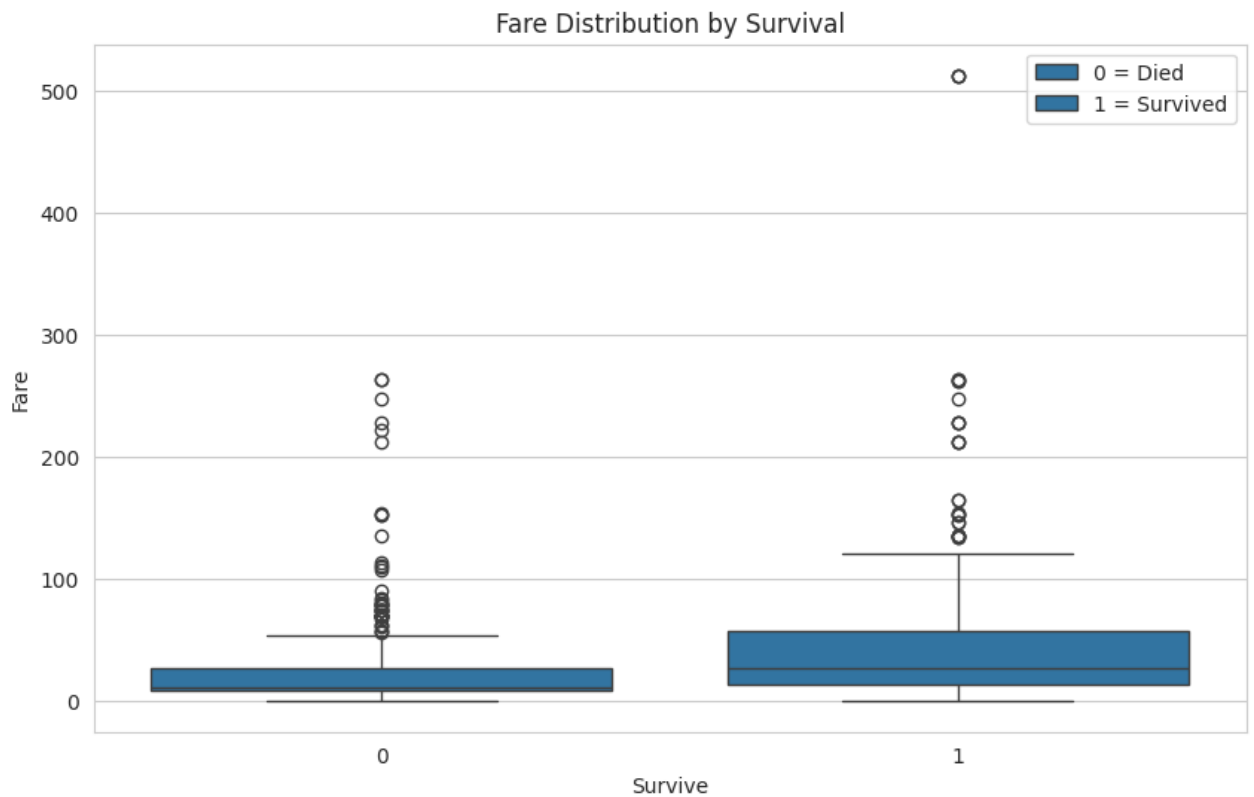
```
# --- ENTER YOUR CODE HERE ---
plt.figure(figsize=(10, 6))

sns.boxplot(
    x = 'Survived',
    y = 'Fare',
    data = df
```

```
)

plt.title('Fare Distribution by Survival')
plt.legend(['0 = Died', '1 = Survived'])
plt.xlabel('Survive')
plt.ylabel('Fare')
plt.show()

print("\n Insight: Passangers that survive get paid higher fares than
```



Insight: Passangers that survive get paid higher fares than passangers
The plot also contains several high fareoutliers.



Instructions: Answer the following questions in this markdown cell.

- 1. What is the primary goal of Exploratory Data Analysis (EDA)?**
- 2. Based on the plots in this lab, what kind of person had the best chance of surviving the Titanic? (Describe them in terms of class, gender, and age).**
- 3. Why is it important to visualize data instead of just looking at summary statistics? What can a plot show you that a number like 'mean' or 'count' can't?**

[ENTER YOUR ANSWERS HERE]

1. The primary goal of the EDA which stands for Exploratory Data Analysis is being able to understand any data set before you start building a model for it. It's almost like your creating a summary for the statistics given and being able to visualize any type of patterns, relationships, or anything unusual happening in the data given.

2. The kind of person that had the best chance of surviving the Titanic is apparently the first class female as shown in visualization 2 and 3. But also seeing that younger children have a higher chance in survival since they seem to possibly have priority.

3. It is important to visualize data instead of just looking at summary statistics because if we just visualize the statistics we could be possibly wrong in what we thought or could make a simple miscalculation or input. Visualizing helps see proper patterns, connections, and relationships and could also hide differences in certain clusters of groups just like in the experiment 2.

